Four Target-Selector Modes for the Safe MPC Filter

Repository Implementation Note

March 26, 2026

1 Overview

The safe MPC filter in this repository uses an offset-free augmented model

$$z_k = \begin{bmatrix} x_k \\ d_k \end{bmatrix}, \qquad z_{k+1} = \begin{bmatrix} A & B_d \\ 0 & I \end{bmatrix} z_k + \begin{bmatrix} B \\ 0 \end{bmatrix} u_k, \qquad y_k = Cx_k + C_d d_k, \tag{1}$$

with the current augmented estimate $\hat{z}_k = [\hat{x}_k; \hat{d}_k]$. The target-selector layer computes a steady package

$$(x_s, u_s, d_s, y_s, y_{c,s}, r_s)$$

used by the downstream MPC objective, Lyapunov center, and safety-correction QCQP. The present implementation supports four selector modes behind one common entrypoint

`prepare_filter_target(..., selector_mode, config, prev_target, H)`.

2 Shared Notation

Let H select controlled outputs so that

$$y_{c,s} = Hy_s, \qquad r_s \in \mathbb{R}^{n_c}.$$

The tightened input and output bounds are

$$u_\ell^t = u_\ell + u_{\text{tight}}, \qquad u_h^t = u_h - u_{\text{tight}},$$

$$y_\ell^t = y_\ell + y_{\text{tight}}, \qquad y_h^t = y_h - y_{\text{tight}}.$$

The steady-state dynamics are written as

$$(I - A)x_s - Bu_s - B_d d_s = 0.$$

Soft output bounds are implemented with nonnegative slack variables and quadratic penalties. Warm start uses the previous accepted target $(x_s^{\text{prev}}, u_s^{\text{prev}}, d_s^{\text{prev}})$ when available.

3 Mode 0: Current Exact/Fallback Frozen-d

Mode 0 preserves the existing two-stage Rawlings-style selector. The disturbance is frozen at

$$d_s = \hat{d}_k.$$

Stage 1 solves

$$\begin{aligned}
\min_{x_s,u_s} \quad & (u_s - u_{\text{nom}})^\top R_u (u_s - u_{\text{nom}}) + x_s^\top Q_x x_s + \Phi_y^{\text{soft}}(y_s) && (2)\\
\text{s.t.} \quad & (I - A)x_s - Bu_s - B_d \hat{d}_k = 0, && (3)\\
& y_s = Cx_s + C_d \hat{d}_k, && (4)\\
& y_{c,s} = Hy_s = y_{sp}, && (5)\\
& u_\ell^t \le u_s \le u_h^t, && (6)
\end{aligned}$$

plus optional output bounds. If Stage 1 is rejected, Stage 2 solves

$$\begin{aligned}
\min_{x_s,u_s} \quad & (y_{c,s} - y_{sp})^\top T_y (y_{c,s} - y_{sp}) + (u_s - u_{\text{nom}})^\top R_u (u_s - u_{\text{nom}}) && (7)\\
& + x_s^\top Q_x x_s + (x_s - x_s^{\text{prev}})^\top Q_{\Delta x} (x_s - x_s^{\text{prev}}) + (u_s - u_s^{\text{prev}})^\top R_{\Delta u} (u_s - u_s^{\text{prev}})\\
& + \Phi_y^{\text{soft}}(y_s) && (8)
\end{aligned}$$

under the same steady-state and bound constraints. In this mode

$$r_s = y_{c,s}.$$

4 Mode 1: Free-Disturbance Prior

Mode 1 treats the pseudo steady disturbance as an optimization variable, not a frozen truth:

$$d_s \text{ is optimized and penalized around } \hat{d}_k.$$

The single-stage problem is

$$\begin{aligned}
\min_{x_s,u_s,d_s} \quad & \|r_s - y_{sp}\|_{Q_r}^2 + \|u_s - u_{\text{nom}}\|_{R_u}^2 + \|x_s - x_s^{\text{prev}}\|_{Q_{\Delta x}}^2 && (9)\\
& + \|u_s - u_s^{\text{prev}}\|_{R_{\Delta u}}^2 + \|d_s - \hat{d}_k\|_{Q_d}^2 + \Phi_y^{\text{soft}}(y_s) && (10)
\end{aligned}$$

subject to

$$\begin{aligned}
(I - A)x_s - Bu_s - B_d d_s &= 0, && (11)\\
y_s &= Cx_s + C_d d_s, && (12)\\
y_{c,s} &= Hy_s, && (13)\\
r_s &= y_{c,s}, && (14)\\
u_\ell^t \le u_s &\le u_h^t. && (15)
\end{aligned}$$

Optionally,

$$\|d_s - \hat{d}_k\|_\infty \le \Delta_d.$$

This mode is designed for the case where the disturbance estimate is a good bias prior but not a trustworthy steady-state coordinate.

5 Mode 2: Compromised Reference

Mode 2 keeps the disturbance frozen but explicitly promotes the admissible steady controlled output to the official target. The problem is

$$\min_{x_s,u_s} \quad \|r_s - y_{sp}\|_{Q_r}^2 + \|u_s - u_{\text{nom}}\|_{R_u}^2 + \|x_s - x_s^{\text{prev}}\|_{Q_{\Delta x}}^2 \tag{16}$$

$$+ \|u_s - u_s^{\text{prev}}\|_{R_{\Delta u}}^2 + \Phi_y^{\text{soft}}(y_s) \tag{17}$$

subject to

$$(I - A)x_s - Bu_s - B_d\hat{d}_k = 0, \tag{18}$$

$$y_s = Cx_s + C_d\hat{d}_k, \tag{19}$$

$$y_{c,s} = Hy_s, \tag{20}$$

$$r_s = y_{c,s}, \tag{21}$$

$$u_\ell^t \le u_s \le u_h^t. \tag{22}$$

The downstream rule in this repository is: when this mode is selected, the upstream MPC candidate, the QCQP output term, and the backup MPC all use r_s as their tracking target whenever dimensions are compatible.

6 Mode 3: Single-Stage Robust SSTP

Mode 3 removes the exact/fallback switch and always solves one softened steady-state target problem. In the present implementation, d_s is frozen at $\hat{d}_k$ and both the steady-state equation and the controlled-output reference are softened:

$$\min_{x_s,u_s,\epsilon_x,\epsilon_y} \quad \|r_s - y_{sp}\|_{Q_r}^2 + \|u_s - u_{\text{nom}}\|_{R_u}^2 + \|x_s - x_s^{\text{prev}}\|_{Q_{\Delta x}}^2 \tag{23}$$

$$+ \|u_s - u_s^{\text{prev}}\|_{R_{\Delta u}}^2 + \rho_x\|\epsilon_x\|_1 + \rho_y\|\epsilon_y\|_1 + \Phi_y^{\text{soft}}(y_s) \tag{24}$$

subject to

$$(I - A)x_s - Bu_s - B_d\hat{d}_k = \epsilon_x, \tag{25}$$

$$y_s = Cx_s + C_d\hat{d}_k, \tag{26}$$

$$y_{c,s} = Hy_s, \tag{27}$$

$$r_s = y_{c,s} + \epsilon_y, \tag{28}$$

$$u_\ell^t \le u_s \le u_h^t. \tag{29}$$

Large ρ_x and ρ_y keep the problem close to a true steady-state target calculation while avoiding the binary exact/fallback switch.

7 Downstream Consistency Rules

The safe filter now uses the selector result through an *effective target*:

- current target if the current solve succeeded;
- otherwise the last valid target if the backup policy is `last_valid`;

- otherwise no target.

The tracking target used by the upstream MPC and QCQP is mode-consistent:

Mode	Default downstream target	Disturbance treatment
Mode 0	current policy (`raw` or admissible)	frozen at $\hat{d}_k$
Mode 1	selector reference if available	optimized with prior
Mode 2	forced selector reference r_s	frozen at $\hat{d}_k$
Mode 3	selector reference r_s by default	frozen at $\hat{d}_k$

8 Optimization Methods

All four modes are convex and are solved with CVXPY through a common solver-preference sequence. The code resets variable values, applies initial values from the previous accepted target when available, and tries the configured solver list in order. Accepted statuses are currently

$$\{\texttt{optimal}, \texttt{optimal_inaccurate}\}.$$

Mode 0 uses an explicit two-stage solve. Modes 1–3 use a single-stage convex problem.

9 Implementation Mapping

The code mapping is:

- `Lyapunov/target_selector.py`
  - `prepare_filter_target(...)`: common API and mode dispatch
  - `build_target_selector_config(...)`: defaults and override merge
  - `compute_ss_target_refined_rawlings(...)`: preserved Mode 0 core
  - `_compute_target_mode2_compromised_reference(...)`
  - `_compute_target_mode1_free_disturbance_prior(...)`
  - `_compute_target_mode3_single_stage_robust_sstp(...)`
- `Simulation/run_mpc_lyapunov.py` and `Simulation/run_rl_lyapunov.py`
  - build selector config once per rollout
  - select the downstream tracking target consistently by mode
  - propagate selector diagnostics into the debug stream
- `Lyapunov/safety_debug.py`
  - stores `selector_mode`, `r_s`, the disturbance-deviation metric $d_s - \hat{d}$, and the decomposition
  - saves plots for Cx_s, $C_d d_s$, and y_s

10 Advantages, Risks, and Expected Failure Modes

Mode 0 is the baseline for reproducibility, but it can spend most of its life in fallback under estimator drift. **Mode 1** is more flexible when $\hat{d}_k$ is not a trustworthy steady-state disturbance, but it can blur state-versus-disturbance identifiability. **Mode 2** removes target mismatch across layers, but if the selector itself is poor then the entire loop consistently tracks a poor admissible target. **Mode 3** removes explicit stage switching, but overly small ρ_x or ρ_y can turn the steady target into a numerically convenient but physically weak surrogate.

11 References

References

[1] J. B. Rawlings, D. Q. Mayne, and M. Diehl, *Model Predictive Control: Theory, Computation, and Design*, 2nd ed., Nob Hill Publishing, 2020.

[2] K. R. Muske and T. A. Badgwell, "Disturbance modeling for offset-free linear model predictive control," *Journal of Process Control*, 2002.

[3] J. Kuntz, *Towards a Turnkey Model Predictive Controller*, PhD thesis, University of California, Santa Barbara, 2024.

[4] X. Yan et al., "A predictive control framework with guaranteed safety," *Automatica*, 2023.

[5] D. Kassmann, T. A. Badgwell, and R. B. Hawkins, "Robust steady-state target calculation for model predictive control," *AIChE Journal*, 2000.